

Ahmed Ikram | ML & AI Engineering

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Specialises in end-to-end ML pipeline engineering, MLOps infrastructure, and LLM-powered application development - having independently built an anomaly detection engine achieving 99.8% $\pm 0.1\%$ StratifiedKFold CV accuracy and an Agentic RAG assistant cutting inference latency from 25-50s to 10-20s, with all training runs versioned in MLflow on AWS. Applies ML rigorously: leakage-safe feature engineering, cross-validated benchmarking, cross-encoder reranking, and LLM-as-a-judge evaluation - validated across 670 tests spanning unit, ML, RAG, and security tiers. Seeking an ML Engineering, AI Engineering, or MLOps post-graduation role to contribute production-grade model pipelines and LLM system design.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, XGBoost, scikit-learn, MLflow, LangChain, ChromaDB, Ollama, Kafka, Redis, PostgreSQL, Docker, AWS | Mar 2026 - May 2026

- Led backend and ML engineering on a 7-person Agile team for NCR Atleos; independently built the 3-layer detection engine, Agentic RAG assistant, full MLOps infrastructure, and a 670-test validation suite across 10 tiers - all built post-submission significantly improving the team-delivered project
- Engineered a 3-layer anomaly detection engine running every 30s: ML Ensemble (XGBoost + Isolation Forest) on 49 hand-engineered features achieving 99.8% $\pm 0.1\%$ StratifiedKFold CV accuracy across 8 classes and 97.3% IF precision via grid search and F1-threshold calibration; Z-Score layer detects distribution shift via rolling 20-window Z-scores; Heuristic layer provides a 7-detector deterministic fallback - all runs versioned in MLflow v3 on AWS (RDS backend + S3 artifact store) with champion aliases
- Built an Agentic RAG diagnostic assistant over ChromaDB + Ollama with cross-encoder reranking (+5-15% retrieval relevance), self-consistency scoring (3 parallel samples, 3-gram Jaccard similarity), reflexion self-critique, and 4-signal confidence fusion; parallelising self-consistency cut latency from ~25-50s to ~10-20s

StockLens FinTech Spending & Investment App | React Native Expo, TypeScript, Node.js, Python, FastAPI, scikit-learn | Oct 2025 - Dec 2025

- Implemented an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI alongside the OCR receipt pipeline

Haggis Species Classification & Predictive Modelling | Python, scikit-learn, XGBoost, Matplotlib, Seaborn, Pandas, NumPy, Jupyter Notebook | Nov 2025 - Dec 2025

- Engineered and validated 2 novel ratio-based features that emerged as top-3 predictors across all models; prevented data leakage using sklearn Pipelines with ColumnTransformer and RobustScaler throughout
- Trained and compared 7 classifiers including XGBoost, Random Forest, and Logistic Regression achieving ~90% accuracy; tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering to validate natural groupings against ground-truth labels; built a Linear Regression model predicting body mass ($R^2=0.756$), validating assumptions via residual diagnostics

DevSync Project Tracker with GitHub Integration | React, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, CI/CD | Jan 2025 - Apr 2025

- Deployed production AWS infrastructure: ECS Fargate + ALB + RDS private subnet + S3 + CloudFront + ACM; rolling deploys tagged by commit SHA for immutable rollback; OIDC federation scoped to main branch with no static credentials stored or transmitted anywhere in the pipeline, repository or task definitions
- 1,446 automated tests (517 Pytest backend, 929 Jest + RTL + Cypress frontend) at 85%/85% line coverage hard-gate every PR - any failure aborts deployment before any AWS step runs or is even triggered

PROJECTS

CineMatch AI / ML Movie Recommendation System | *Python, scikit-learn, Pandas, Flask, NumPy* | *May 2025 - July 2025*

- Implemented three recommendation approaches - collaborative filtering, content-based filtering, and a hybrid model - on the MovieLens dataset; collaborative filtering achieved ~78% hit rate and ~0.22 Precision@10
- Systematically compared all three approaches across hit rate, Precision@10, and cold-start handling - hybrid model outperformed standalone methods by combining user similarity signals with item content features, with each strategy independently swappable via dependency injection for controlled benchmarking

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Bash
ML & AI	scikit-learn, XGBoost, Pandas, NumPy, MLflow, Artifact Stores, Auto-retrain Pipelines
LLM Engineering	LangChain, ChromaDB, Ollama, Vector Search, Evaluation Pipelines, RAG
Data & Backend	Apache Airflow, ETL/ELT, SQLite, PostgreSQL, NoSQL, MySQL, REST APIs, FastAPI, Apache Kafka
Cloud & Tools	AWS (EC2, S3, RDS, IAM, SageMaker), Docker, Redis, Azure, GitHub Actions, Linux
Visualisation	Jupyter Notebook, Matplotlib, Seaborn

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>

EXPERIENCE

Beaulys Mini Market , Dundee, Scotland <i>Retail Assistant</i>	<i>September 2023 - Present</i>
• Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption	
Zulus Peri Peri Grill , Dundee, Scotland <i>Front of House Worker</i>	<i>June 2021 - September 2023</i>
• Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure	

TRANSFERRABLE SKILLS

Model Evaluation & Benchmarking Benchmark models rigorously - cross-validation, leakage prevention, held-out evaluation sets, and LLM-as-judge pipelines - not just headline accuracy scores	Experimental & Hypothesis-Driven Thinking Frame ML problems as falsifiable hypotheses; validate with held-out sets and residual diagnostics before drawing conclusions
LLM Integration & RAG Architecture Design and evaluate end-to-end LLM-powered systems - chunking strategy, embedding selection, vector retrieval, and generation quality as first-class architectural concerns	Data Pipeline Engineering for ML Build production-grade data pipelines that feed ML systems reliably - ingestion, transformation, synthetic data generation, and anomaly injection
Self-directed Learning Independently acquired AWS ML and Azure AI certifications alongside full-time study; consistently extends technical depth beyond coursework	Research Methodology Apply structured research practices, controlled evaluation, feature validation, and statistical testing, to ensure findings generalise beyond training data
Technical Communication Present model results, retrieval quality metrics, and system trade-offs clearly to both technical peers and non-technical stakeholders	Cross-functional Collaboration Delivered ML and AI components within a multi-disciplinary Agile team; experienced coordinating across data engineering, frontend, and analysis workstreams